Risolvere problemi di programmazione intera

Algoritmi di ottimizzazione [266SM]

Lorenzo Castelli

26 Marzo 2026

UNIVERSITÀ DEGLI STUDI DI TRIESTE

Enumeration. All feasible solutions are identified and the best one is picked up. It may not be practically viable. For instance, to solve the TSP in a complete graph with n nodes there are $(n-1)!$ feasible tours. Hence,

n	$n!$
10	3.6×10^6
100	9.33×10^{157}
1000	4.02×10^{2567}

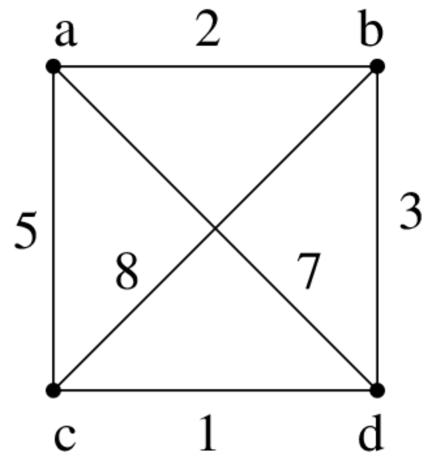
Better ideas are needed.

The Travelling Salesman Problem

- We are given a set on nodes $V = \{1, \dots, n\}$ (e.g., cities) and a set of arcs $\mathcal{A}$.
- Arcs represent ordered pairs of cities between which direct travel is possible.
- For $(i, j) \in \mathcal{A}, c_{ij}$ is the direct travel time from city i to city j.
- The TSP aims at finding a tour, starting at city 1, that
  a) visits each other city exactly once and then returns to city 1
  b) takes the least total travel time

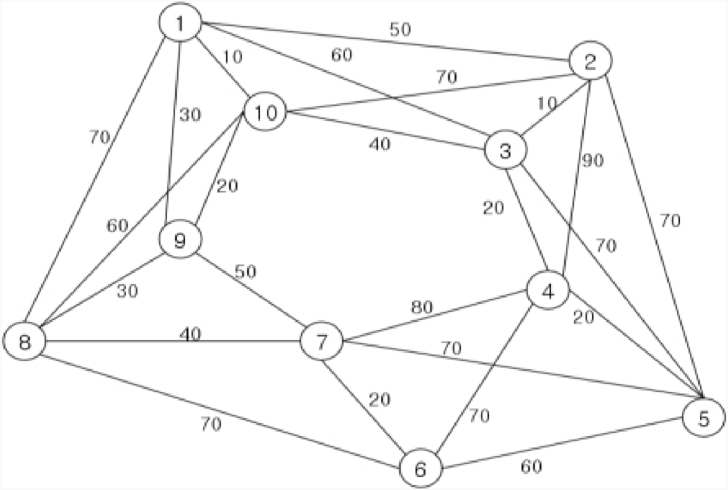
The Travelling Salesman Problem (TSP)

A tour that visits all nodes exactly once is called Hamiltonian tour.
The TSP identifies the Hamiltonian tour of minimum cost.

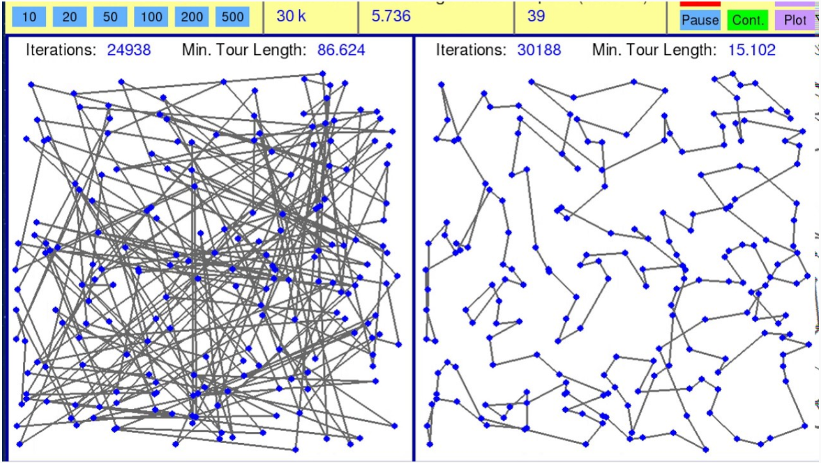


Three tours: A-B-D-C-A: 11; A-D-B-C-A: 23; A-D-C-B-A: 18.

TSP - maybe not too easy



TSP - it's difficult!!



TSP - Formulation

- Decision Variables

$$x_{ij} = \begin{cases} 1 & \text{if } j \text{ immediately follows } i \text{ on the tour} \\ 0 & \text{otherwise} \end{cases}$$

  Hence $x \in \{0,1\}^{|\mathcal{A}|}$

- Objective function

$$\min \sum_{(i,j)\in\mathcal{A}} c_{ij} x_{ij}$$

TSP - Constraint formulation

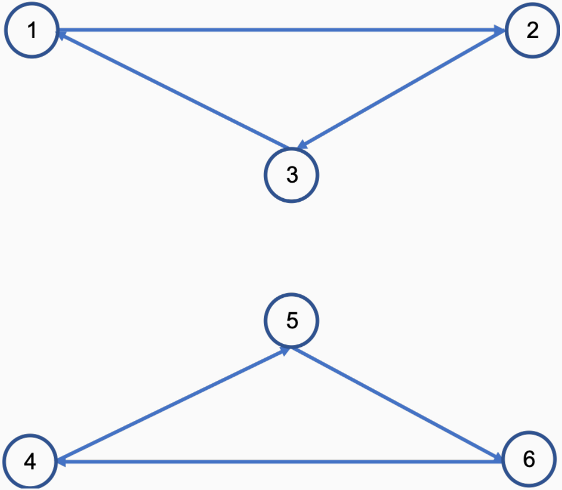
Each city is entered and left exactly once

$$\sum_{i:(i,j)\in\mathcal{A}} x_{ij} = 1 \text{ for } j \in V \tag{1}$$

$$\sum_{j:(i,j)\in\mathcal{A}} x_{ij} = 1 \text{ for } i \in V \tag{2}$$

However, constraints (1) and (2) are not sufficient to define tours since they are also satisfied by subtours.

TSP - Subtours



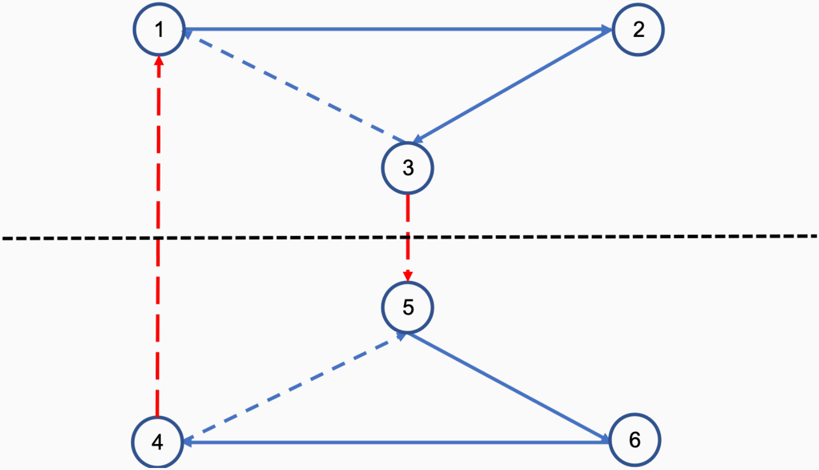
TSP - Subtour elimination (i)

In any tour there must be an arc that goes from $\{1, 2, 3\}$ to $\{4, 5, 6\}$ and an arc that goes from $\{4, 5, 6\}$ to $\{1, 2, 3\}$. In general, for any $U \subset V$ with $2 \leq |U| \leq |V| - 2$, constraints

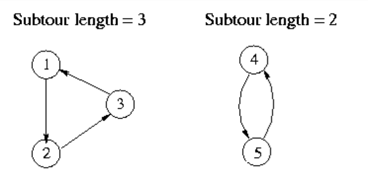
$$\sum_{\{(i,j)\in\mathcal{A}: i\in U, j\in V\setminus U\}} x_{ij} \geq 1 \tag{3}$$

are satisfied by all tours, but every subtour violates at least one of them.

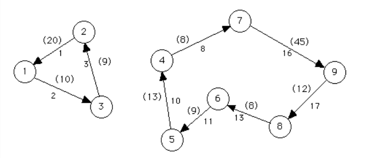
TSP - Subtour elimination



TSP - Subtours

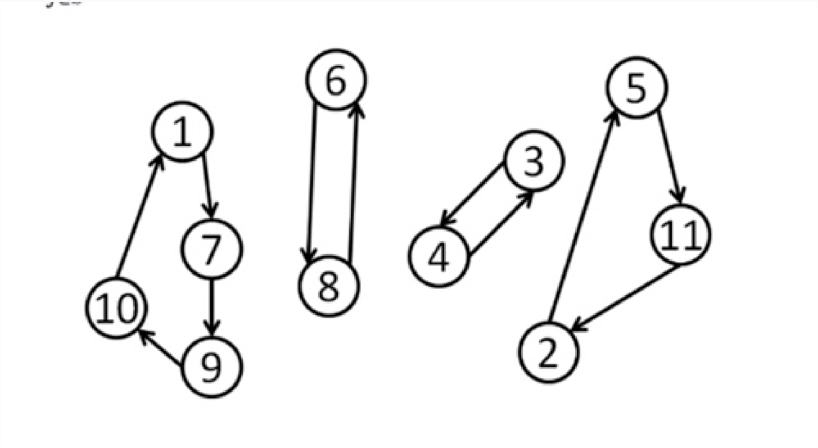


(a) 5 nodes

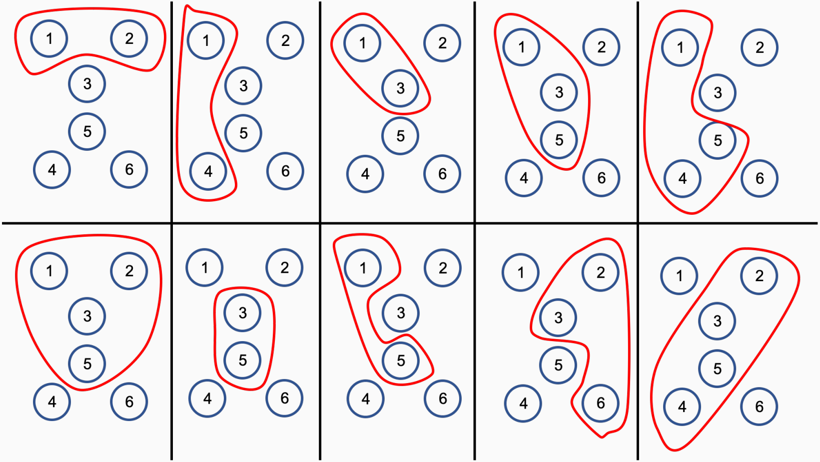


(b) 9 nodes

TSP - Subtours



TSP - Too many ways to choose U



An alternative way to eliminate subtours is to introduce constraints

$$\sum_{\{(i,j)\in\mathcal{A}:i\in U,j\in U\}} x_{ij} \leq |U| - 1 \ \forall U \subset V : 2 \leq |U| \leq |V| - 2 \qquad (4)$$

But again we need a constraint for each $U \subset V$ such that $2 \leq |U| \leq |V| - 2$.

In both (3) and (4) the number of constraints is nearly $2^{|V|}$!!!

$$\frac{1}{2}\left[\binom{|V|}{2} + \binom{|V|}{3} + \cdots + \binom{|V|}{|V|-2}\right]$$

TSP - Formulation

$$\min \sum_{(i,j)\in\mathcal{A}} c_{ij}x_{ij}$$

$$\sum_{i:(i,j)\in\mathcal{A}} x_{ij} = 1 \text{ for } j \in V$$

$$\sum_{j:(i,j)\in\mathcal{A}} x_{ij} = 1 \text{ for } i \in V$$

$$\sum_{\{(i,j)\in\mathcal{A}:i\in U,j\in V\setminus U\}} x_{ij} \geq 1 \qquad \forall U \subset V : 2 \leq |U| \leq |V| - 2$$

OR

$$\sum_{\{(i,j)\in\mathcal{A}:i\in U,j\in U\}} x_{ij} \leq |U| - 1 \qquad \forall U \subset V : 2 \leq |U| \leq |V| - 2$$

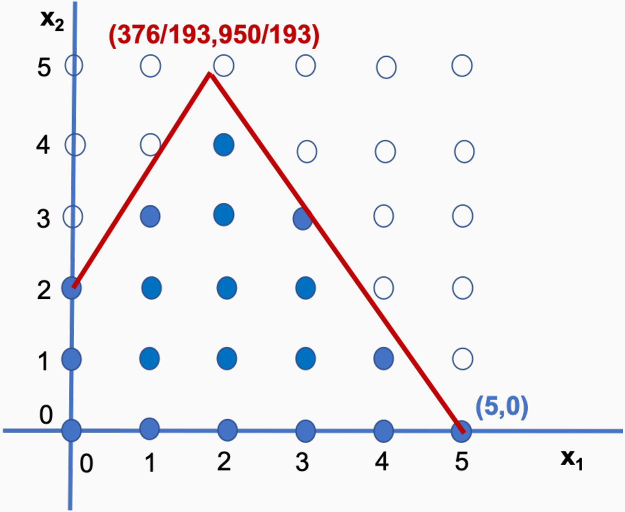
$$x \in \{0,1\}^{|\mathcal{A}|}$$

Disregarding variables' integrality constraints. Consider the following problem:

$$\begin{aligned}\max Z = & 1.00x_1 + 0.64x_2 \\ & 50x_1 + 31x_2 \leq 250 \\ & 3x_1 - 2x_2 \geq -4 \\ & x_1, x_2 \geq 0 \text{ and integer.}\end{aligned}$$

- The optimal integer solution is $(5, 0)$
- The optimal solution without considering variables' integrality constraints is $(376/193, 950/193) = (1.948, 4.922)$

How to solve an IP?



How to solve an IP?

Disregarding variables' integrality constraints. Why not to round up and/or down the linear solution?

- The upper integer part $(\lceil 1.948 \rceil, \lceil 4.922 \rceil) = (2, 5)$ is NOT FEASIBLE (the first constraint is violated).
- The lower integer part $(\lfloor 1.948 \rfloor, \lfloor 4.922 \rfloor) = (1, 4)$ is NOT FEASIBLE (the second constraint is violated).
- The mixed choice $(\lfloor 1.948 \rfloor, \lceil 4.922 \rceil) = (1, 5)$ is NOT FEASIBLE (the second constraint is violated).
- The mixed choice $(\lceil 1.948 \rceil, \lfloor 4.922 \rfloor) = (2, 4)$ is feasible but NOT OPTIMAL: $Z(2, 4) = 4.56$, whereas $Z(5, 0) = 5$.

In addition, no rounding gives the values $(5, 0)$.

In conclusion, the linear solution appears to be useless to find the integer solution.

Thank you for your attention